

STA 322 Project 1: Characteristics of Trinity College Faculty

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Introduction

This project uses a probability sample to estimate characteristics of faculty in Duke’s Trinity College of Arts & Sciences. We answer four questions, each with a point estimate and a 95% confidence interval:

1. What fraction of faculty are women?
2. What fraction of faculty have at least one grant that is ongoing in 2026?
3. Among faculty with a Ph.D., what is the average number of years since the degree?
4. Do women and men differ in the fraction who are full professors?

The sampling frame is built from departmental web pages, and the data for each sampled faculty member come from their Scholars@Duke profile. Although it might be possible to collect information on every faculty member, the purpose of the project is to practice survey sampling, so we drew a probability sample and use design-based estimation. Question 1 is the primary target, so we chose the design to estimate a proportion precisely while keeping data collection manageable for a two-person team.

Target Population and Sampling Frame

The target population is everyone listed on the faculty web pages of Trinity College’s 36 departments and programs as of October 1, 2026. For each unit we used the “primary faculty” page where one exists, because it lists faculty whose main appointment is in that department. Some units do not post a primary-faculty page; for those (e.g., English, Theater Studies, Music, Religious Studies) we used the full faculty listing. The population therefore includes tenure-track faculty, lecturers, instructors, research and practice faculty, ROTC cadre, and any emeriti whom a department still lists.

Because some departments list all of their faculty rather than only primary faculty, a faculty member with a joint appointment can appear on more than one roster. Removing these duplicates would require matching every name across all 36 rosters before sampling. We therefore retained every listing and treated each faculty member appearing on a department’s page as a member of that department. The practical consequence is that a person listed twice has two chances of selection; we flag such cases in our data and discuss them as a limitation.

We built the frame by counting each roster by hand and recording, for each unit, the number of faculty listed. This gives a frame of $N = 866$ faculty across 36 units. Only these counts were needed to design the sample; the names themselves were looked up only for sampled positions.

Survey Design

Overview

We used **stratified simple random sampling without replacement with proportional allocation**, with a total sample size of $n = 100$ (above the suggested minimum of 75). The population is

partitioned into $H = 27$ strata: departments, with a few very small units pooled (explained below). Within each stratum h we selected a simple random sample of n_h roster positions out of the N_h faculty listed.

The sample-size calculation, the full allocation table, and the derivation of the weights and variance estimator are given in the Appendix.

Defining the strata

Each department is its own stratum unless it is too small to receive at least two sampled faculty under proportional allocation. With proportional allocation, stratum h receives nN_h/N units before rounding, so a department needs

$$\frac{nN_h}{N} \geq 2 \iff N_h \geq \frac{2N}{n} = \frac{2 \times 866}{100} \approx 17.3$$

faculty to get two sampled people on its own. Units below this size were pooled with related units:

- **Classical & Religious Studies:** Classical Studies, Religious Studies
- **Dance & Theater:** Dance, Theater Studies
- **Education & Gender Studies:** Education; Gender, Sexuality & Feminist Studies
- **Small Languages:** German Studies, Romance Studies, Slavic & Eurasian Studies, International Comparative Studies
- **Military & PE:** Air Force ROTC, Army ROTC, Navy ROTC, Health, Wellness & Physical Education

We pooled related fields so that each pooled stratum is still fairly homogeneous, which preserves most of the benefit of stratification.

Implementing the design in R

The `survey` package needs four pieces of information about each sampled faculty member to reproduce this design:

- `ids = ~1`: there is no clustering; each faculty member is selected individually.
- `strata = ~stratum`: the stratum h the person was sampled from, so variances are computed within strata and then combined.
- `weights = ~weight`: the design weight $1/\pi_i = N_h/n_h$.
- `fpc = ~N_h`: the stratum population size N_h , from which `survey` computes n_h/N_h and applies the finite population correction $1 - n_h/N_h$.

```
library(survey)
des <- svydesign(ids = ~1, strata = ~stratum, weights = ~weight,
               fpc = ~N_h, data = dat)
```

As a check, the estimated total of a variable equal to 1 for everyone, `svytotal(~one, des)`, equals $\sum_{i \in S} 1/\pi_i = 866$, the frame size. The estimator and variance formula that `survey` computes from these inputs are derived in Appendix C, and the full code is in Appendix D. Results for each question are in the Results and Analysis section.

Rationale for Our Design Choices

This section explains why we chose to stratify by department and why we allocated the sample proportionally.

Why stratify by department

We stratified by department mainly for precision. The share of women and the prevalence of grants differ sharply across fields, and when the stratum means $\bar{Y}_h$ are spread out, the within-stratum variances S_h^2 are small relative to S^2 , so stratified sampling has smaller variance than a simple random sample of the same size. Stratifying also guarantees that every department appears in the sample, so no field can be missed by chance. Finally, it was convenient: the frame is already organized by department, so the stratum sizes N_h came directly from counting each roster, and sampling within a stratum only requires that department's page.

Why proportional allocation

We allocated the sample proportionally, $n_h \approx nN_h/N$, rounding so the total is exactly 100. Proportional allocation guarantees $\text{Var}(\hat{t}_{prop}) \leq \text{Var}(\hat{t}_{srs})$, because it removes the between-strata variation from the variance. Unlike optimal allocation, which needs the unknown S_h and is tuned to a single variable, it requires only the stratum sizes N_h and serves all four questions equally. It is also nearly self-weighting: every faculty member's selection probability is $\pi_i = n_h/N_h \approx n/N$, so the weights $1/\pi_i$ are nearly equal.

The resulting n_h , inclusion probabilities, and weights for all 27 strata are listed in Appendix B.

Data Collection

Drawing the sample

We drew the sample in R with `set.seed(322)` so it can be reproduced exactly. In each stratum h we selected n_h distinct roster positions at random from $1, \dots, N_h$. In pooled strata, positions run through the member departments in a fixed order (for example, in Dance & Theater, positions 1 to 7 are Dance and 8 to 28 are Theater Studies), and each position was mapped back to a department and a position on that department's page. The sampled person is the name at that position, counting down the page in the order shown on the website. The sampling code is reproduced in Appendix D.

We divided collection by stratum: Janet collected the strata from African & African American Studies through History, and Amy collected the remaining strata.

Variables and coding rules

For each sampled person we recorded the following from their Scholars@Duke profile, using rules we agreed on in advance so that both collectors coded the same way:

- **Sex (Question 1).** Coded from pronouns in the profile biography where given, and otherwise from first name and profile photo, as the assignment directs.
- **Ongoing grant in 2026 (Question 2).** Coded 1 if any grant listed on the profile has an award period that overlaps calendar year 2026, regardless of the person's role (principal investigator, co-investigator, or mentor on a training grant). Fellowships and gifts, which Scholars lists separately from grants, were not counted.
- **Ph.D. status and year (Question 3).** From the Education section. A D.Phil. was counted as a

Ph.D.; an M.F.A., M.Ed., M.A., or other degree was not. Years since Ph.D. is 2026 – Ph.D. year.

- **Rank (Question 4).** We recorded the full title and coded *full professor* as 1 for any title at the rank of Professor, including named and distinguished chairs, Professor of the Practice, Research Professor, and emeriti at that rank. Assistant and associate professors, lecturers, instructors, and other titles were coded 0.

Missing data and frame issues

- **Missing information.** A few profiles were incomplete. One faculty member had disabled most profile features, and three others had no Ph.D. year listed. For these four, we took the Ph.D. year from the person’s personal website. One person’s grant status could not be determined (their personal website lists a grant awarded in 2022 with no end date), so it was recorded as missing. Missing values are excluded from the estimate for that question only.
- **Emeriti.** A few sampled faculty are emeriti who are still listed on a department roster. They belong to our target population as defined, so we kept them rather than replacing them, since replacing sampled units would change the selection probabilities.
- **Roster changes.** The Gender, Sexuality & Feminist Studies page listed 8 faculty when we built the frame and 9 when we collected data. We used the person at the sampled position on the current page and treat this as a minor frame error.
- **Multiple listings.** Several sampled faculty also appear on another department’s roster (for example, faculty with joint appointments in African & African American Studies or Theater Studies). As permitted for this project, we did not adjust their weights. Because these faculty had a higher chance of selection, any characteristic associated with joint appointments may be slightly overrepresented in our estimates.

Data Cleaning and Missing Data

The final data set has 100 sampled faculty. Before analysis we checked that every record had a valid stratum, that coded variables took only the values 0 and 1, and that a Ph.D. year was present only for faculty coded as having a Ph.D. We recorded 86 faculty with a Ph.D. and 14 without (for example M.F.A., M.Ed. or M.A. as the highest degree). Missing values were handled as follows:

- **Ph.D. year** is blank for the 14 faculty without a Ph.D., which is expected, and for no one who has one.
- **Ongoing grant** is missing for 1 faculty member whose profile has the grant information turned off.
- **Sex, Ph.D. status and rank** have no missing values.

Rather than impute, each estimate uses the sampled faculty who have the variable observed. With only one missing value in the analysis, the choice has no practical effect on any estimate.

Results and Analysis

All estimates below use the design weights and the stratified Horvitz-Thompson estimator with the finite population correction (Appendix C). Proportions are estimated with `svyciprop`, and the mean in Question 3 with `svymean` on the subset of Ph.D. holders.

Table 1: Design-based estimates with 95% confidence intervals ($n = 100$).

Question	Estimate	95% CI
1. Women	44.4%	36.2% to 52.9%
2. Ongoing grant in 2026	29.1%	21.5% to 38.2%
3. Mean years since Ph.D. (Ph.D. holders)	23.6 years	20.8 to 26.4 years
4a. Full professor, all faculty	52.3%	42.7% to 61.6%
4b. Full professor, women	52.2%	37.3% to 67.2%
4c. Full professor, men	52.3%	40.2% to 64.4%
4d. Difference in full professors (women - men)	-0.1%	-19.4% to 19.3%

Question 1: Fraction of faculty who are women

We estimate that 44.4% of Trinity faculty are women (95% CI: 36.2% to 52.9%). The sample contained 44 women and 56 men. The interval includes 50%, so the data do not rule out an even split, but they are more consistent with women being a minority of the faculty.

Question 2: Ongoing grants in 2026

We estimate that 29.1% of faculty have at least one grant ongoing in 2026 (95% CI: 21.5% to 38.2%). Most faculty therefore do not have an active grant. We counted a grant if its award period overlapped 2026, regardless of the faculty member's role, so training grants where the person is a mentor are included. A stricter definition (principal investigator only) would give a lower fraction.

Question 3: Years since Ph.D.

Among Ph.D. holders, the average time since the degree is 23.6 years (95% CI: 20.8 to 26.4). We estimate that about 86% of faculty hold a Ph.D. (95% CI: 79.6% to 90.0%), and the sample's Ph.D. years range from 0 to 57 years ago. The range is large because the faculty include recent hires and senior professors.

Question 4: Do women and men differ in rank?

Our "full professor" code groups ranks that differ in meaning (distinguished and endowed chairs, Professors of the Practice, and emeriti with the rank). We estimate that 52.3% of faculty are full professors overall (95% CI: 42.7% to 61.6%). For women the estimate is 52.2% and for men 52.3%. The estimated difference is -0.1% (95% CI: -19.4 to 19.3), and the Rao-Scott test gives $p = 0.99$. We therefore find no evidence of a difference in the share of full professors between women and men. This comparison does not adjust for years since the Ph.D., so any real gap in rank may be hidden or explained by seniority.

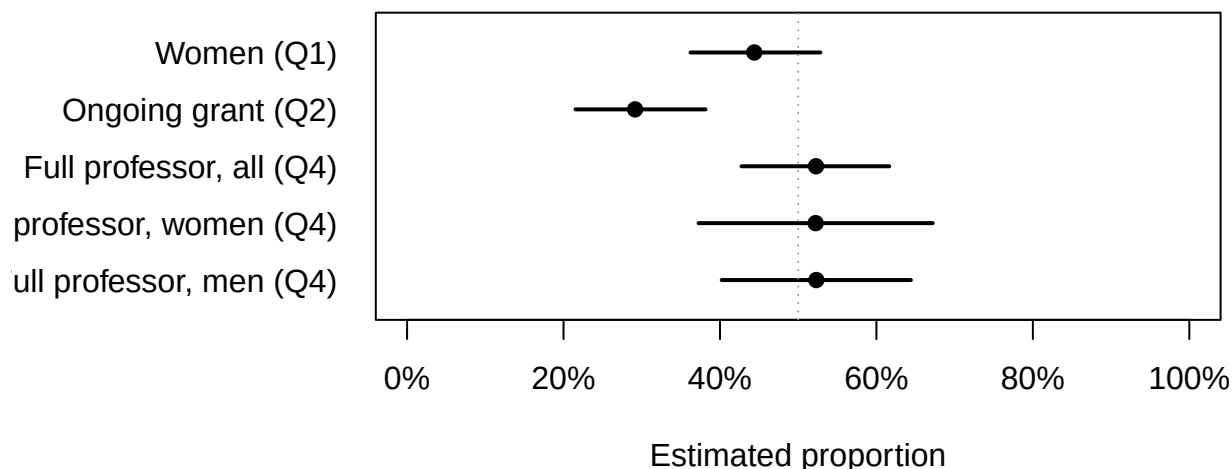


Figure 1: Estimated proportions with 95% confidence intervals.

Limitations

- **Measurement of sex.** Sex was coded using pronouns in the biography where available and otherwise first name and photo. A few profiles state no sex identifiers, so some codes are inferences that may not match how those faculty identify. Misclassification would bias Question 1 slightly and would also affect the Question 4 comparison.
- **Frame and duplicate listings.** Faculty with joint appointments can appear on more than one roster. Those faculty had more than one chance of selection but received the weight of the single department they were drawn from, so estimates are slightly biased toward their characteristics.
- **Design.** Several strata have only two or three sampled faculty, so the stratum-level variance estimates are unstable even though the overall variance is sound. Pooling unrelated small departments, though necessary, gives those strata less meaningful interpretations.
- **Missing data.** The one missing grant value was excluded, not imputed. This has negligible effect here but assumes that faculty who disable their profile are not systematically different.

Sensitivity analysis

Table 2: Sensitivity: excluding the 17 emeriti and double-listed faculty (the remaining 83 are kept in the design)

Question	All 100	Without emeriti and double-listed
1. Women	44.4%	45.1%
2. Ongoing grant	29.1%	30.2%
3. Mean years since Ph.D.	23.6	22.5
4a. Full professor, all	52.3%	51.0%

As a check on the two judgment calls above (not included in the conclusion), the table recomputes each estimate after removing the emeriti and double-listed faculty from the sample, keeping the rest of the design unchanged. The largest change is in Question 3, where the average falls by about a year, as expected since emeriti are among the most senior faculty and have the longest time since their Ph.D.

Conclusion

We used a stratified simple random sample of 100 faculty, allocated in proportion to department size, to estimate characteristics of Trinity College faculty. We estimate that about 44% of faculty are women (95% CI: 36.2% to 52.9%), that about 29% have a grant ongoing in 2026 (95% CI: 21.5% to 38.2%), and that Ph.D. holders earned their degree an average of 24 years ago (95% CI: 20.8 to 26.4). For our additional question, the share of full professors is nearly identical for women (52%) and men (52%), with no statistically detectable difference, although the sample is too small to rule out a moderate one.

The main limits are the width of the intervals at $n = 100$, the hand-coded sex and grant variables, and duplicate roster listings. With more time we would increase the sample, run a pilot to estimate department-level variability for optimal allocation, and measure rank more finely and adjust for years since the Ph.D. to learn whether women and men differ in rank once seniority is accounted for.

Appendix

A. Sample size

We chose $n = 100$, comfortably above the suggested 75 and manageable to collect in a few hours each. We sample $n/N = 100/866 \approx 0.115$ of the population. Ignoring the gain from stratification, the worst-case margin of error for a proportion (when the proportion is 0.5, so $S^2 \approx 0.25$) is

$$1.96\sqrt{\left(1 - \frac{n}{N}\right) \frac{S^2}{n}} = 1.96\sqrt{\left(1 - \frac{100}{866}\right) \frac{0.25}{100}} \approx 0.092,$$

about ± 9 percentage points. Stratification should make the actual intervals somewhat narrower.

B. Allocation and weights

Because we take a simple random sample within each stratum, faculty member i in stratum h has inclusion probability

$$\pi_i = \Pr(I_i = 1) = \frac{n_h}{N_h},$$

and their **design weight** is the inverse, $1/\pi_i = N_h/n_h$. The weight is the number of population faculty that each sampled person represents. The weights within a stratum sum to N_h , so across all strata $\sum_{i \in S} 1/\pi_i = N = 866$.

Under proportional allocation, $n_h \approx nN_h/N$. Because nN_h/N is rarely a whole number, we used largest-remainder rounding: round every n_h down, then give the remaining units, one each, to the strata with the largest fractional parts, so the total is exactly 100. The code below computes the strata, the allocation, and the weights from our roster counts.

```
frame <- read.csv("faculty_counts.csv", stringsAsFactors = FALSE)
N <- sum(frame$total_faculty)
n <- 100

# Pool units too small to receive 2 sampled faculty on their own
pool <- list(
  "Military & PE (pooled)" = c("Air Force ROTC", "Army ROTC", "Navy ROTC",
                              "Health, Wellness & Physical Education"),
  "Small languages (pooled)" = c("International Comparative Studies",
                                 "Slavic & Eurasian Studies", "Romance Studies",
                                 "German Studies"),
  "Dance & Theater (pooled)" = c("Dance", "Theater Studies"),
  "Education & Gender Studies (pooled)" = c("Education",
                                             "Gender, Sexuality & Feminist Studies"),
  "Classical & Religious Studies (pooled)" = c("Classical Studies",
                                                "Religious Studies")
)
frame$stratum <- frame$department
for (s in names(pool)) frame$stratum[frame$department %in% pool[[s]]] <- s

strata <- aggregate(total_faculty ~ stratum, data = frame, FUN = sum)
names(strata)[2] <- "N_h"
```

```

# Proportional allocation with largest-remainder rounding
strata$exact <- n * strata$N_h / N
strata$n_h <- floor(strata$exact)
short <- n - sum(strata$n_h)
bump <- order(strata$exact - strata$n_h, decreasing = TRUE)[seq_len(short)]
strata$n_h[bump] <- strata$n_h[bump] + 1

# Inclusion probabilities and design weights
strata$pi <- strata$n_h / strata$N_h
strata$weight <- strata$N_h / strata$n_h

stopifnot(sum(strata$n_h) == n, all(strata$n_h >= 2),
          isTRUE(all.equal(sum(strata$n_h * strata$weight), N)))

```

Table 3: Proportional allocation and design weights by stratum

Stratum	N_h	nN_h/N	n_h	$\pi_i = n_h/N_h$	$1/\pi_i = N_h/n_h$
African & African American Studies	18	2.08	2	0.111	9.00
Art, Art History & Visual Studies	27	3.12	3	0.111	9.00
Asian & Middle Eastern Studies	45	5.20	5	0.111	9.00
Biology	61	7.04	7	0.115	8.71
Chemistry	27	3.12	3	0.111	9.00
Classical & Religious Studies (pooled)	27	3.12	3	0.111	9.00
Computer Science	38	4.39	4	0.105	9.50
Cultural Anthropology	17	1.96	2	0.118	8.50
Dance & Theater (pooled)	28	3.23	3	0.107	9.33
Economics	60	6.93	7	0.117	8.57
Education & Gender Studies (pooled)	20	2.31	2	0.100	10.00
English	33	3.81	4	0.121	8.25
Evolutionary Anthropology	17	1.96	2	0.118	8.50
History	27	3.12	3	0.111	9.00
Linguistics	20	2.31	2	0.100	10.00
Literature	17	1.96	2	0.118	8.50
Mathematics	58	6.70	7	0.121	8.29
Military & PE (pooled)	23	2.66	3	0.130	7.67
Music	65	7.51	7	0.108	9.29
Philosophy	18	2.08	2	0.111	9.00
Physics	40	4.62	5	0.125	8.00
Political Science	32	3.70	4	0.125	8.00
Psychology & Neuroscience	41	4.73	5	0.122	8.20
Small languages (pooled)	25	2.89	3	0.120	8.33
Sociology	25	2.89	3	0.120	8.33

Stratum	N_h	nN_h/N	n_h	$\pi_i = n_h/N_h$	$1/\pi_i = N_h/n_h$
Statistical Science	26	3.00	3	0.115	8.67
Thompson Writing Program	31	3.58	4	0.129	7.75

The 27 strata have n_h between 2 and 7, and the weights range from 7.67 to 10, close to the overall $N/n = 8.66$. Rounding is the only source of the differences: strata rounded down (e.g., Education & Gender Studies, $2.31 \rightarrow 2$) have slightly larger weights, and strata rounded up (e.g., Military & PE, $2.66 \rightarrow 3$) have slightly smaller ones.

C. Estimation and variance

Plugging the inclusion probabilities into the Horvitz-Thompson estimator gives the stratified estimators of the population total and mean:

$$\hat{t}_y = \sum_{i \in S} \frac{y_{hi}}{\pi_i} = \sum_{h=1}^H N_h \bar{y}_h, \quad \bar{y}_{HT} = \frac{\hat{t}_y}{N} = \sum_{h=1}^H \frac{N_h}{N} \bar{y}_h.$$

Each stratum's sample mean $\bar{y}_h$ is weighted by that stratum's share of the population. For a 0/1 variable such as sex, $\bar{y}_h$ is the sample proportion in stratum h , and $\bar{y}_{HT}$ estimates the population proportion. Because $\hat{t}_y$ is a Horvitz-Thompson estimator, it is unbiased.

Because we take separate simple random samples in each stratum, the $\bar{y}_h$ are independent, and the variance of $\bar{y}_{HT}$ is

$$\text{Var}(\bar{y}_{HT}) = \sum_{h=1}^H \frac{N_h^2}{N^2} \left(1 - \frac{n_h}{N_h}\right) \frac{S_h^2}{n_h}, \quad S_h^2 = \sum_{i=1}^{N_h} \frac{(y_{hi} - \bar{Y}_h)^2}{N_h - 1}.$$

We estimate each S_h^2 with the sample variance s_h^2 , giving

$$\widehat{\text{Var}}(\bar{y}_{HT}) = \sum_{h=1}^H \frac{N_h^2}{N^2} \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h},$$

and the 95% confidence interval $\bar{y}_{HT} \pm 1.96 \sqrt{\widehat{\text{Var}}(\bar{y}_{HT})}$. For a 0/1 variable, $s_h^2 = n_h \bar{y}_h(1 - \bar{y}_h)/(n_h - 1)$.

Two parts of this formula shaped the design. The factor $1 - n_h/N_h$ is the **finite population correction**. We sample about 11.5% of each stratum without replacement, so this factor reduces the variance by about that much, and leaving it out would overstate our uncertainty. The divisor $n_h - 1$ in s_h^2 is why every stratum needs at least two sampled units.

Question 3 (average years since Ph.D. among Ph.D. holders) is a **domain estimate**: the number of Ph.D. holders in each stratum is not known in advance, so the estimate is a ratio of two estimated totals. We compute it by restricting the full design object to Ph.D. holders, which keeps the correct stratified variance.

D. Code

Drawing the sample

This code drew the roster positions used for data collection. It is shown, not re-run, so the report does not overwrite the sample file.

```

set.seed(322)
rows <- list()
for (i in seq_len(nrow(strata))) {
  s <- strata$stratum[i]
  members <- frame[frame$stratum == s, ]
  pos <- sort(sample(strata$N_h[i], strata$n_h[i]))
  ends <- cumsum(members$total_faculty)
  starts <- ends - members$total_faculty
  idx <- findInterval(pos - 1, ends) + 1
  rows[[i]] <- data.frame(stratum = s, department = members$department[idx],
                        roster_position = pos - starts[idx])
}
sample_ids <- do.call(rbind, rows)
sample_ids$weight <- strata$weight[match(sample_ids$stratum, strata$stratum)]
write.csv(sample_ids, "sample_ids.csv", row.names = FALSE)

```

Analysis

The code that builds the design object and computes every estimate in the Results and Analysis section:

```

library(survey)
knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE)

# Rebuild the strata exactly as in analysis.Rmd (pooled small units)
frame <- read.csv("faculty_counts.csv", stringsAsFactors = FALSE)
pool <- list(
  "Military & PE (pooled)" = c("Air Force ROTC", "Army ROTC", "Navy ROTC",
                              "Health, Wellness & Physical Education"),
  "Small languages (pooled)" = c("International Comparative Studies",
                                 "Slavic & Eurasian Studies", "Romance Studies",
                                 "German Studies"),
  "Dance & Theater (pooled)" = c("Dance", "Theater Studies"),
  "Education & Gender Studies (pooled)" = c("Education",
                                             "Gender, Sexuality & Feminist Studies"),
  "Classical & Religious Studies (pooled)" = c("Classical Studies",
                                              "Religious Studies"))

frame$stratum <- frame$department
for (s in names(pool)) frame$stratum[frame$department %in% pool[[s]]] <- s
N_h <- tapply(frame$total_faculty, frame$stratum, sum)

dat <- read.csv("data.csv", na.strings = c("NA", ""), stringsAsFactors = FALSE)
stopifnot(all(dat$stratum %in% names(N_h)))
n_h <- table(dat$stratum)
dat$N_h <- as.numeric(N_h[dat$stratum])
dat$weight <- dat$N_h / as.numeric(n_h[dat$stratum])
dat$years_since_phd <- ifelse(dat$has_phd %in% 1, 2026 - dat$phd_year, NA)

```

```

dat$sex <- factor(dat$female, levels = c(0, 1), labels = c("Men", "Women"))
dat$emeritus <- grepl("meritus", dat$rank_title) | grepl("^Emeritus", dat$notes)
dat$double_listed <- grepl("roster", dat$notes, ignore.case = TRUE) & !dat$emeritus
des <- svydesign(ids = ~1, strata = ~stratum, weights = ~weight,
               fpc = ~N_h, data = dat)

# helpers for inline numbers
pct <- function(x, d = 1) sprintf("%.1f%%", d, 100 * as.numeric(x))
ci_pct <- function(ci) sprintf("%.1f%% to %.1f%%", 100 * ci[1], 100 * ci[2])

# estimates
d1 <- subset(des, !is.na(female))
q1 <- svyciprop(~female, d1); q1_ci <- confint(q1)
d2 <- subset(des, !is.na(ongoing_grant_2026))
q2 <- svyciprop(~ongoing_grant_2026, d2); q2_ci <- confint(q2)
d3 <- subset(des, has_phd %in% 1 & !is.na(years_since_phd))
q3 <- svymean(~years_since_phd, d3); q3_ci <- confint(q3)
q3b <- svyciprop(~I(has_phd == 1), subset(des, !is.na(has_phd)))
q3b_ci <- confint(q3b)
d4 <- subset(des, !is.na(female) & !is.na(full_professor))
q4 <- svyciprop(~full_professor, d4); q4_ci <- confint(q4)
q4_by <- svyby(~full_professor, ~sex, d4, svymean, vartype = c("se", "ci"),
              covmat = TRUE)
q4_diff <- svycontrast(q4_by, c(-1, 1))
q4_test <- svychisq(~full_professor + sex, d4)
q4_diff_ci <- as.numeric(q4_diff) + c(-1, 1) * 1.96 * as.numeric(SE(q4_diff))

# counts for the text
n_missing <- colSums(is.na(dat[, c("female", "ongoing_grant_2026", "has_phd",
                                   "phd_year", "full_professor")]))

n_phd <- sum(dat$has_phd %in% 1)
n_nophd <- sum(dat$has_phd %in% 0)
n_women <- sum(dat$female %in% 1)
n_men <- sum(dat$female %in% 0)
n_emer <- sum(dat$emeritus)
n_dbl <- sum(dat$double_listed)

# sensitivity: drop emeriti and double-listed faculty from the sample
des2 <- subset(des, !emeritus & !double_listed)
s1 <- svyciprop(~female, subset(des2, !is.na(female)))
s2 <- svyciprop(~ongoing_grant_2026, subset(des2, !is.na(ongoing_grant_2026)))
s3 <- svymean(~years_since_phd, subset(des2, has_phd %in% 1 & !is.na(years_since_phd)))
s4 <- svyciprop(~full_professor, des2)
res <- data.frame(
  Question = c("1. Women", "2. Ongoing grant in 2026",
               "3. Mean years since Ph.D. (Ph.D. holders)",

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      "4a. Full professor, all faculty",
      "4b. Full professor, women",
      "4c. Full professor, men",
      "4d. Difference in full professors (women - men)"),
  Estimate = c(sprintf("%.1f%%", 100 * q1), sprintf("%.1f%%", 100 * q2),
    sprintf("%.1f years", coef(q3)), sprintf("%.1f%%", 100 * q4),
    sprintf("%.1f%%", 100 * q4_by$full_professor[2]),
    sprintf("%.1f%%", 100 * q4_by$full_professor[1]),
    sprintf("%.1f%%", 100 * as.numeric(q4_diff))),
  `95% CI` = c(ci_pct(q1_ci), ci_pct(q2_ci),
    sprintf("%.1f to %.1f years", q3_ci[1], q3_ci[2]),
    ci_pct(q4_ci),
    ci_pct(c(q4_by$ci_l[2], q4_by$ci_u[2])),
    ci_pct(c(q4_by$ci_l[1], q4_by$ci_u[1])),
    sprintf("%.1f%% to %.1f%%", 100 * q4_diff_ci[1], 100 * q4_diff_ci[2])),
  check.names = FALSE)
knitr::kable(res, align = "lrr", caption = "Design-based estimates with 95% confidence intervals")
est <- c(as.numeric(q1), as.numeric(q2), as.numeric(q4),
  q4_by$full_professor[2], q4_by$full_professor[1])
lo <- c(q1_ci[1], q2_ci[1], q4_ci[1], q4_by$ci_l[2], q4_by$ci_l[1])
hi <- c(q1_ci[2], q2_ci[2], q4_ci[2], q4_by$ci_u[2], q4_by$ci_u[1])
labs <- c("Women (Q1)", "Ongoing grant (Q2)", "Full professor, all (Q4)", "Full professor, women")
par(mar = c(4, 9.5, 1, 1))
y <- rev(seq_along(est))
plot(est, y, xlim = c(0, 1), ylim = c(0.5, length(est) + 0.5), pch = 19,
  yaxt = "n", xaxt = "n", xlab = "Estimated proportion", ylab = "")
segments(lo, y, hi, y, lwd = 2)
axis(1, at = seq(0, 1, 0.2), labels = paste0(seq(0, 100, 20), "%"))
axis(2, at = y, labels = labs, las = 1, tick = FALSE)
abline(v = 0.5, lty = 3, col = "grey60")
sens <- data.frame(
  Question = c("1. Women", "2. Ongoing grant", "3. Mean years since Ph.D.",
    "4a. Full professor, all"),
  `All 100` = c(sprintf("%.1f%%", 100 * q1), sprintf("%.1f%%", 100 * q2),
    sprintf("%.1f", coef(q3)), sprintf("%.1f%%", 100 * q4)),
  `Without emeriti and double-listed` = c(sprintf("%.1f%%", 100 * s1),
    sprintf("%.1f%%", 100 * s2),
    sprintf("%.1f", coef(s3)),
    sprintf("%.1f%%", 100 * s4)),
  check.names = FALSE)
knitr::kable(sens, align = "lrr", caption = sprintf("Sensitivity: excluding the %d emeriti and c"))

```